M21_202601

2026-01-27

Contents

Setup **1**
0.1 A. RT — Words (primary behavioral model) . . . 1
0.2 B. RT — Nonwords (primary behavioral model) . . . 2
0.3 C. ERP — Words . . . 3
0.3.1 C1. N250 — Words . . . 3
0.3.2 C2. N400 — Words . . . 4
0.4 D. ERP — Nonwords . . . 5
0.4.1 D1. N250 — Nonwords . . . 5
0.4.1.1 Complexity × OS collapsed over Family Size (N250 nonwords) . . . 6
0.4.1.2 Complexity × LP collapsed over Family Size (N250 nonwords) . . . 7
0.4.2 D2. N400 — Nonwords (strongest ERP result) . . . 8
0.4.2.1 N400 (nonwords): (Complex - Simple) difference scores as a function of continuous Language Proficiency (LP), averaged over Family Size . . . 10

Setup

Load libraries

1. Set `ggplot2` parameters

0.1 A. RT — Words (primary behavioral model)

Question: Does orthographic sensitivity predict efficiency, controlling lexical familiarity?

```
library(lme4)

rt_w <- read.csv("rt_words_frq.csv") %>%
  filter(correct == 1) %>%
  mutate(
    SubjID = factor(SubjID),
    Item   = factor(target),
    logRT  = log(response_time),
    OS     = as.numeric(Dim.2),
    LP     = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )


m_rt_words <- lmer(
  logRT ~ zLogBF + zLogFS + OS + LP +
    zLogBF:OS + zLogFS:OS +
    (1 + zLogBF + zLogFS | SubjID) +
    (1 | Item),
  data = rt_w,
  control = lmerControl(optimizer = "bobyqa")
)

summary(m_rt_words)
```

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: logRT ~ zLogBF + zLogFS + OS + LP + zLogBF:OS + zLogFS:OS + (1 +      zLogBF + zLogFS | SubjID) + (1 | Item)
||    Data: rt_w
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: -3520
```

```
||
|| Scaled residuals:
||     Min      1Q  Median      3Q     Max
|| -9.8804 -0.6674 -0.0668  0.5975  5.0420
||
|| Random effects:
||  Groups   Name        Variance  Std.Dev. Corr
||  Item     (Intercept) 1.807e-03 0.042505
||  SubjID   (Intercept) 1.770e-02 0.133042
||           zLogBF      2.832e-05 0.005322 -0.59
||           zLogFS      5.019e-05 0.007085 -0.08 -0.76
||  Residual             2.955e-02 0.171914
|| Number of obs: 5847, groups:  Item, 100; SubjID, 67
||
|| Fixed effects:
||               Estimate Std. Error         df t value Pr(>|t|)
|| (Intercept)   6.390222   0.016957  72.480776 376.853  < 2e-16 ***
|| zLogBF       -0.017515   0.004742 101.848722  -3.694 0.000358 ***
|| zLogFS       -0.012862   0.004940  93.201727  -2.604 0.010734 *
|| OS           -0.034670   0.018724  63.877129  -1.852 0.068701 .
|| LP            0.012792   0.012199  63.971487   1.049 0.298316
|| zLogBF:OS    -0.002448   0.002639 191.846848  -0.928 0.354767
|| zLogFS:OS     0.001207   0.002735  78.224041   0.441 0.660288
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||           (Intr) zLogBF zLogFS OS     LP     zLBF:O
|| zLogBF    -0.069
|| zLogFS    -0.010 -0.121
|| OS         0.012 -0.002  0.001
|| LP         0.008  0.000  0.000  0.008
|| zLogBF:OS -0.003  0.024 -0.003 -0.165  0.001
|| zLogFS:OS  0.001 -0.003  0.004 -0.026  0.000 -0.174
|| optimizer (bobyqa) convergence code: 0 (OK)
|| boundary (singular) fit: see help('isSingular')
```

Report: main effects + OS moderation only. LP is included but not emphasized.

0.2 B. RT — Nonwords (primary behavioral model)

Question: Does morphological complexity slow decisions, and is this cost modulated by OS?

```r
rt_nw <- read.csv("rt_nwords_frq.csv") %>%
  filter(correct == 1) %>%
  mutate(
    SubjID = factor(SubjID),
    ItemID = factor(ItemID),
    complexity = factor(complexity),  # SIMP / COMP
    logRT  = log(response_time),
    OS     = as.numeric(Dim.2),
    LP     = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

m_rt_nw <- lmer(
  logRT ~ complexity + zLogBF + zLogFS + OS + LP +
    complexity:OS +
    (1 + complexity | SubjID) +
    (1 | ItemID),
  data = rt_nw,
  control = lmerControl(optimizer = "bobyqa")
)

summary(m_rt_nw)
```

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + complexity:OS +      (1 + complexity | SubjID) + (1 | ItemID)
||    Data: rt_nw
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: -4157.6
||
|| Scaled residuals:
||     Min      1Q  Median      3Q     Max
|| -4.2588 -0.6639 -0.0233  0.6631  3.8762
||
|| Random effects:
```

```
||  Groups   Name           Variance  Std.Dev. Corr
||  ItemID   (Intercept)    0.0013407 0.03662
||  SubjID   (Intercept)    0.0168964 0.12999
||           complexitySIMP 0.0005785 0.02405  0.08
||  Residual                0.0214368 0.14641
|| Number of obs: 4607, groups:  ItemID, 100; SubjID, 67
||
|| Fixed effects:
||                    Estimate Std. Error        df t value Pr(>|t|)
|| (Intercept)        6.578947   0.016671 68.019550 394.636  < 2e-16 ***
|| complexitySIMP    -0.048876   0.005379 61.581469  -9.086 5.62e-13 ***
|| zLogBF             0.015466   0.005080 98.230857   3.045  0.00299 **
|| zLogFS            -0.006353   0.005137 96.653766  -1.237  0.21922
|| OS                -0.041664   0.018536 61.520383  -2.248  0.02819 *
|| LP                 0.003281   0.012232 62.907937   0.268  0.78938
|| complexitySIMP:OS  0.002288   0.006117 59.948618   0.374  0.70971
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||             (Intr) cmSIMP zLogBF zLogFS OS     LP
|| cmplxtySIMP -0.092
|| zLogBF      -0.005  0.005
|| zLogFS      -0.002 -0.011 -0.576
|| OS           0.009  0.011 -0.001 -0.001
|| LP           0.001  0.015 -0.002  0.000  0.010
|| cmplSIMP:OS  0.011 -0.040  0.000  0.003 -0.091 -0.006
```

0.3 C. ERP — Words

Run **separately by time window** (this is important).

0.3.1 C1. N250 — Words

Question: Does orthographic sensitivity modulate early sensitivity to morphological family structure?

```r
erp_w <- read.csv("erp_dmg_words.csv") %>%
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# ERP words
contrasts(erp_w$family_size) <- contr.sum(2)

m_w_N250 <- lmer(
  value ~ family_size*OS + family_size*LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_w, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)

summary(m_w_N250)
```

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * OS + family_size * LP + (1 | SubjID) +      (1 | SubjID:chlabel)
||    Data: subset(erp_w, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15032.9
||
|| Scaled residuals:
||     Min      1Q  Median      3Q     Max
|| -8.4814 -0.4012 -0.0214  0.3759 12.9242
||
|| Random effects:
||  Groups         Name        Variance Std.Dev.
||  SubjID:chlabel (Intercept) 4.626    2.151
||  SubjID         (Intercept) 2.918    1.708
||  Residual                   2.752    1.659
|| Number of obs: 3240, groups:  SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||                   Estimate Std. Error         df t value Pr(>|t|)
```

```
|| (Intercept)       -0.75979    0.22921   57.00006  -3.315   0.0016 **
|| family_size1       0.16008    0.02920 1617.00005   5.483 4.84e-08 ***
|| OS                 0.26085    0.25635   57.00006   1.018   0.3132
|| LP                -0.27860    0.18636   57.00006  -1.495   0.1404
|| family_size1:OS    0.05455    0.03265 1617.00005   1.671   0.0950 .
|| family_size1:LP   -0.03623    0.02374 1617.00005  -1.526   0.1272
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||             (Intr) fmly_1 OS     LP     f_1:OS
|| family_siz1  0.000
|| OS          -0.046  0.000
|| LP           0.038  0.000  0.021
|| fmly_sz1:OS  0.000 -0.046  0.000  0.000
|| fmly_sz1:LP  0.000  0.038  0.000  0.000  0.021
```

Follow-up (only if interaction significant):

```r
library(emmeans)
emmeans(m_w_N250, ~ family_size | OS,
        at = list(OS = c(-1, 0, 1)))
```

```
|| OS = -1:
||  family_size emmean    SE   df lower.CL upper.CL
||  Large       -0.900 0.355 58.9   -1.609  -0.1905
||  Small       -1.114 0.355 58.9   -1.824  -0.4051
||
|| OS =  0:
||  family_size emmean    SE   df lower.CL upper.CL
||  Large       -0.585 0.231 58.9   -1.047  -0.1225
||  Small       -0.908 0.231 58.9   -1.370  -0.4461
||
|| OS =  1:
||  family_size emmean    SE   df lower.CL upper.CL
||  Large       -0.269 0.338 58.9   -0.946   0.4080
||  Small       -0.702 0.338 58.9   -1.379  -0.0247
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
```

0.3.2 C2. N400 — Words

Question: Does language proficiency shape semantic integration of family structure?

```r
m_w_N400 <- lmer(
  value ~ family_size*LP + family_size*OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)

summary(m_w_N400)
```

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * LP + family_size * OS + (1 | SubjID) +      (1 | SubjID:chlabel)
||    Data: subset(erp_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15567.8
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -10.9198  -0.3868   0.0081   0.4087   6.1947
||
|| Random effects:
||  Groups         Name        Variance Std.Dev.
||  SubjID:chlabel (Intercept) 5.238    2.289
||  SubjID         (Intercept) 4.575    2.139
||  Residual                   3.303    1.817
|| Number of obs: 3240, groups:  SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||                    Estimate Std. Error         df t value Pr(>|t|)
|| (Intercept)      2.618e-01  2.842e-01  5.700e+01   0.921   0.3608
|| family_size1     2.243e-01  3.199e-02  1.617e+03   7.011 3.46e-12 ***
|| LP               1.231e-03  2.311e-01  5.700e+01   0.005   0.9958
|| OS               6.771e-01  3.179e-01  5.700e+01   2.130   0.0375 *
|| family_size1:LP -3.489e-02  2.601e-02  1.617e+03  -1.342   0.1799
```

```
|| family_size1:OS  4.444e-02  3.578e-02  1.617e+03   1.242   0.2143
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||            (Intr) fmly_1 LP     OS     f_1:LP
|| family_siz1  0.000
|| LP           0.038  0.000
|| OS          -0.046  0.000  0.021
|| fmly_sz1:LP  0.000  0.038  0.000  0.000
|| fmly_sz1:OS  0.000 -0.046  0.000  0.000  0.021
```

0.4 D. ERP — Nonwords

Again, **separate by time window**.

0.4.1 D1. N250 — Nonwords

Question: Is early morpho-orthographic parsing sensitive to complexity and family size?

```r
erp_nw <- read.csv("erp_dmg_nonwords.csv") %>%
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size),
    complexity = factor(complexity),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# ERP nonwords
contrasts(erp_nw$complexity)  <- contr.sum(2)
contrasts(erp_nw$family_size) <- contr.sum(2)

m_nw_N250 <- lmer(
  value ~ complexity*family_size*OS + complexity*family_size*LP +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_nw, time_window == "N250"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_nw_N250)
```

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * family_size * OS + complexity * family_size *      LP + (1 | SubjID) + (1 | SubjID:chlabel)
||    Data: subset(erp_nw, time_window == "N250")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 29955.4
||
|| Scaled residuals:
||     Min      1Q  Median      3Q     Max
|| -7.8223 -0.5071 -0.0036  0.4947 12.2205
||
|| Random effects:
||  Groups         Name        Variance Std.Dev.
||  SubjID:chlabel (Intercept) 3.111    1.764
||  SubjID         (Intercept) 2.163    1.471
||  Residual                   4.067    2.017
|| Number of obs: 6480, groups:  SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||                              Estimate Std. Error         df t value Pr(>|t|)
|| (Intercept)                  -0.66509    0.19682   56.99997  -3.379 0.001317 **
|| complexity1                   0.11631    0.02510 4851.00000   4.635 3.67e-06 ***
|| family_size1                  0.08850    0.02510 4851.00000   3.526 0.000425 ***
|| OS                            0.29808    0.22013   56.99997   1.354 0.181051
|| LP                           -0.20675    0.16003   56.99997  -1.292 0.201577
|| complexity1:family_size1      0.03943    0.02510 4851.00000   1.571 0.116200
|| complexity1:OS               -0.19600    0.02807 4851.00000  -6.982 3.29e-12 ***
|| family_size1:OS               0.03258    0.02807 4851.00000   1.161 0.245896
|| complexity1:LP               -0.17882    0.02041 4851.00000  -8.763  < 2e-16 ***
|| family_size1:LP              -0.02070    0.02041 4851.00000  -1.014 0.310451
|| complexity1:family_size1:OS  -0.06503    0.02807 4851.00000  -2.317 0.020555 *
|| complexity1:family_size1:LP  -0.02455    0.02041 4851.00000  -1.203 0.228944
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
```

```
|| Correlation of Fixed Effects:
||             (Intr) cmplx1 fmly_1 OS     LP     cm1:_1 cm1:OS f_1:OS cm1:LP f_1:LP c1:_1:O
|| complexity1  0.000
|| family_siz1  0.000  0.000
|| OS          -0.046  0.000  0.000
|| LP           0.038  0.000  0.000  0.021
|| cmplxty1:_1  0.000  0.000  0.000  0.000  0.000
|| cmplxty1:OS  0.000 -0.046  0.000  0.000  0.000  0.000
|| fmly_sz1:OS  0.000  0.000 -0.046  0.000  0.000  0.000  0.000
|| cmplxty1:LP  0.000  0.038  0.000  0.000  0.000  0.000  0.021  0.000
|| fmly_sz1:LP  0.000  0.000  0.038  0.000  0.000  0.000  0.000  0.021  0.000
|| cmpl1:_1:OS  0.000  0.000  0.000  0.000  0.000 -0.046  0.000  0.000  0.000  0.000
|| cmpl1:_1:LP  0.000  0.000  0.000  0.000  0.000  0.038  0.000  0.000  0.000  0.000  0.021
```

```r
# Confirm factor levels (optional but good)
with(subset(erp_nw, time_window == "N250"), table(complexity, family_size))
```

0.4.1.1 Complexity × OS collapsed over Family Size (N250 nonwords)

```
||           family_size
|| complexity Large Small
||    Complex  1620  1620
||    Simple   1620  1620
```

```r
# Estimated marginal means of complexity at OS = -1, 0, +1
emm_cx_by_OS <- emmeans(
  m_nw_N250,
  ~ complexity | OS,
  at = list(OS = c(-1, 0, 1)),
  # This is the key: average over family_size
  weights = "proportional"
)
emm_cx_by_OS
```

```
|| OS = -1:
||  complexity emmean    SE  df asymp.LCL asymp.UCL
||  Complex    -0.632 0.304 Inf    -1.229   -0.0356
||  Simple     -1.274 0.304 Inf    -1.871   -0.6775
||
|| OS =  0:
||  complexity emmean    SE  df asymp.LCL asymp.UCL
||  Complex    -0.530 0.198 Inf    -0.919   -0.1416
||  Simple     -0.780 0.198 Inf    -1.169   -0.3915
||
|| OS =  1:
||  complexity emmean    SE  df asymp.LCL asymp.UCL
||  Complex    -0.428 0.291 Inf    -0.998    0.1414
||  Simple     -0.286 0.291 Inf    -0.856    0.2835
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95
```

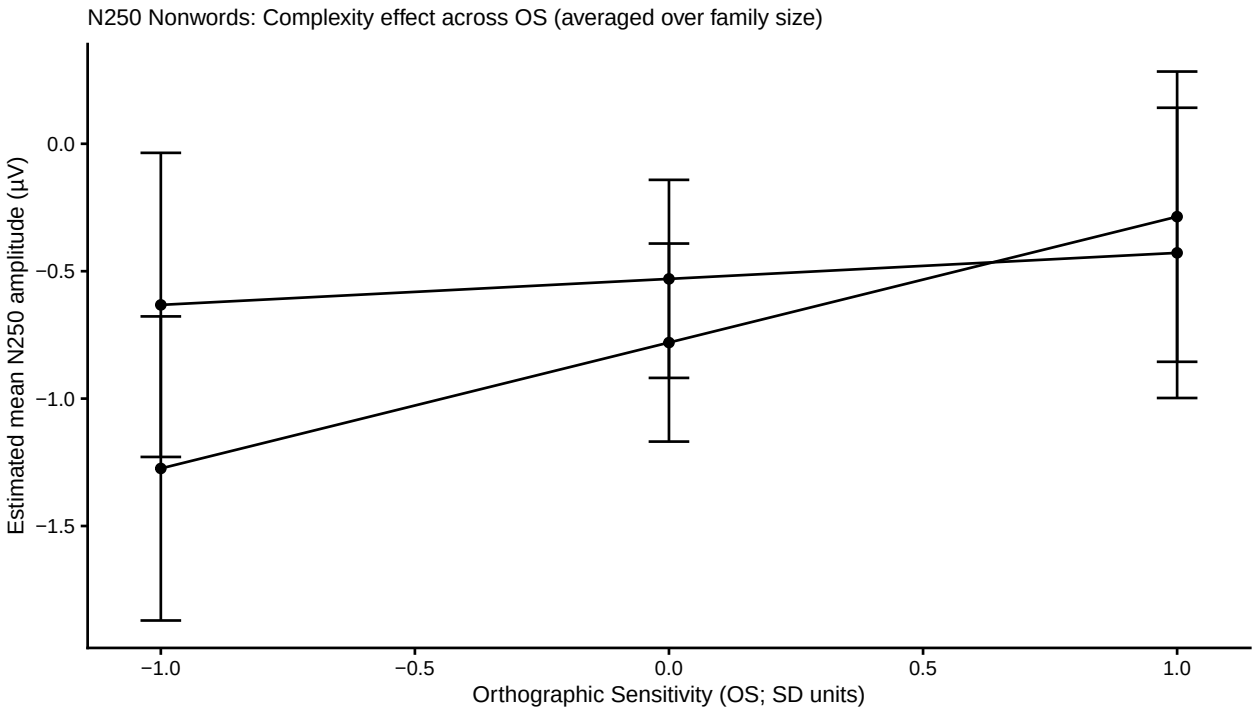
```r
# "Slopes": complexity effect (Complex - Simple) at OS = -1, 0, +1
slopes_cx_OS <- contrast(emm_cx_by_OS, method = "pairwise")  # Complex - Simple within each OS
slopes_cx_OS
```

```
|| OS = -1:
||  contrast         estimate     SE  df z.ratio p.value
||  Complex - Simple    0.642 0.0770 Inf   8.335  <.0001
||
|| OS =  0:
||  contrast         estimate     SE  df z.ratio p.value
||  Complex - Simple    0.250 0.0502 Inf   4.982  <.0001
||
|| OS =  1:
||  contrast         estimate     SE  df z.ratio p.value
||  Complex - Simple   -0.142 0.0735 Inf  -1.933  0.0532
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic
```

```r
# Plot the EMMs (collapsed over family size)
df_emm_OS <- as.data.frame(emm_cx_by_OS)

p_OS <- ggplot(df_emm_OS, aes(x = OS, y = emmean, group = complexity)) +
  geom_line() +
  geom_point() +
  geom_errorbar(aes(ymin = asymp.LCL, ymax = asymp.UCL), width = 0.08) +
  labs(
    x = "Orthographic Sensitivity (OS; SD units)",
```

```
    y = "Estimated mean N250 amplitude (µV)",
    title = "N250 Nonwords: Complexity effect across OS (averaged over family size)"
  )
p_OS
```




```
# Estimated marginal means of complexity at LP = -1, 0, +1
emm_cx_by_LP <- emmeans(
  m_nw_N250,
  ~ complexity | LP,
  at = list(LP = c(-1, 0, 1)),
  weights = "proportional"
)
emm_cx_by_LP
```

0.4.1.2 Complexity × LP collapsed over Family Size (N250 nonwords)

```
|| LP = -1:
||  complexity emmean    SE  df asymp.LCL asymp.UCL
||  Complex    -0.159 0.251 Inf    -0.650     0.332
||  Simple     -0.733 0.251 Inf    -1.224    -0.242
||
|| LP =  0:
||  complexity emmean    SE  df asymp.LCL asymp.UCL
||  Complex    -0.545 0.198 Inf    -0.933    -0.156
||  Simple     -0.761 0.198 Inf    -1.149    -0.372
||
|| LP =  1:
||  complexity emmean    SE  df asymp.LCL asymp.UCL
||  Complex    -0.930 0.260 Inf    -1.441    -0.420
||  Simple     -0.789 0.260 Inf    -1.299    -0.278
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95
```

```
# "Slopes": complexity effect (Complex - Simple) at LP = -1, 0, +1
slopes_cx_LP <- contrast(emm_cx_by_LP, method = "pairwise")  # Complex - Simple within each LP
slopes_cx_LP
```

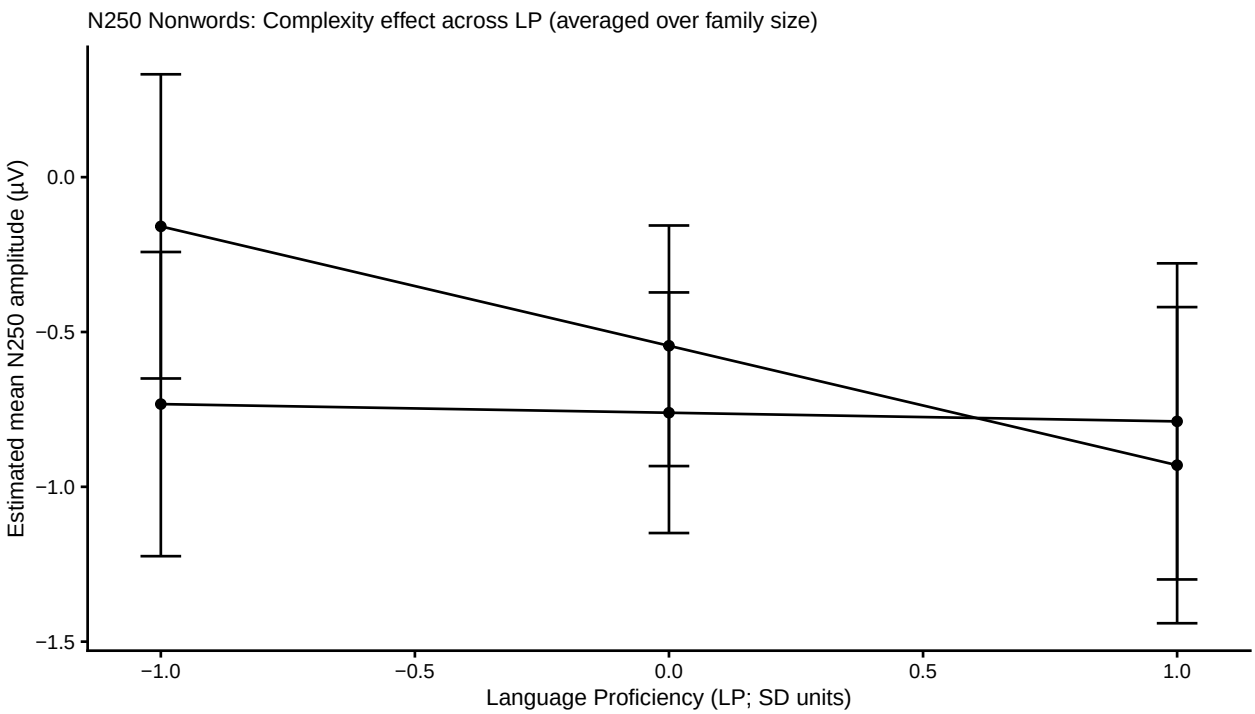
```
|| LP = -1:
||  contrast         estimate     SE  df z.ratio p.value
||  Complex - Simple    0.574 0.0634 Inf   9.052  <.0001
||
|| LP =  0:
||  contrast         estimate     SE  df z.ratio p.value
```

```
||  Complex - Simple    0.216 0.0501 Inf   4.313  <.0001
||
|| LP =  1:
||  contrast         estimate     SE  df z.ratio p.value
||  Complex - Simple   -0.141 0.0659 Inf  -2.146  0.0319
||
|| Results are averaged over the levels of: family_size
|| Degrees-of-freedom method: asymptotic
```

```r
# Plot the EMMs (collapsed over family size)
df_emm_LP <- as.data.frame(emm_cx_by_LP)

p_LP <- ggplot(df_emm_LP, aes(x = LP, y = emmean, group = complexity)) +
  geom_line() +
  geom_point() +
  geom_errorbar(aes(ymin = asymp.LCL, ymax = asymp.UCL), width = 0.08) +
  labs(
    x = "Language Proficiency (LP; SD units)",
    y = "Estimated mean N250 amplitude (µV)",
    title = "N250 Nonwords: Complexity effect across LP (averaged over family size)"
  )

p_LP
```



0.4.2 D2. N400 — Nonwords (strongest ERP result)

Question: Does language proficiency determine how familiarity modulates complexity costs?

```r
m_nw_N400 <- lmer(
  value ~ complexity*family_size*LP + complexity*family_size*OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)

summary(m_nw_N400)
```

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * family_size * LP + complexity * family_size *      OS + (1 | SubjID) + (1 | SubjID:chlabel)
||    Data: subset(erp_nw, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 32991.3
||
```

```
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -14.0373  -0.4804  -0.0160   0.4598  14.1912
||
|| Random effects:
||  Groups         Name        Variance Std.Dev.
||  SubjID:chlabel (Intercept) 6.392    2.528
||  SubjID         (Intercept) 5.568    2.360
||  Residual                   6.100    2.470
|| Number of obs: 6480, groups:  SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||                              Estimate Std. Error         df t value Pr(>|t|)
|| (Intercept)                  -0.45163    0.31311   57.00003  -1.442 0.154666
|| complexity1                   0.23471    0.03074 4851.00002   7.636 2.68e-14 ***
|| family_size1                  0.04236    0.03074 4851.00002   1.378 0.168178
|| LP                            0.01139    0.25458   57.00003   0.045 0.964463
|| OS                            0.38123    0.35019   57.00003   1.089 0.280904
|| complexity1:family_size1      0.11149    0.03074 4851.00002   3.627 0.000290 ***
|| complexity1:LP               -0.12704    0.02499 4851.00002  -5.084 3.84e-07 ***
|| family_size1:LP               0.06110    0.02499 4851.00002   2.445 0.014528 *
|| complexity1:OS               -0.11999    0.03438 4851.00002  -3.490 0.000487 ***
|| family_size1:OS               0.03283    0.03438 4851.00002   0.955 0.339609
|| complexity1:family_size1:LP  -0.05734    0.02499 4851.00002  -2.294 0.021819 *
|| complexity1:family_size1:OS  -0.03566    0.03438 4851.00002  -1.037 0.299619
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||             (Intr) cmplx1 fmly_1 LP     OS     cm1:_1 cm1:LP f_1:LP cm1:OS f_1:OS c1:_1:L
|| complexity1  0.000
|| family_siz1  0.000  0.000
|| LP           0.038  0.000  0.000
|| OS          -0.046  0.000  0.000  0.021
|| cmplxty1:_1  0.000  0.000  0.000  0.000  0.000
|| cmplxty1:LP  0.000  0.038  0.000  0.000  0.000  0.000
|| fmly_sz1:LP  0.000  0.000  0.038  0.000  0.000  0.000  0.000
|| cmplxty1:OS  0.000 -0.046  0.000  0.000  0.000  0.000  0.021  0.000
|| fmly_sz1:OS  0.000  0.000 -0.046  0.000  0.000  0.000  0.000  0.021  0.000
|| cmpl1:_1:LP  0.000  0.000  0.000  0.000  0.000  0.038  0.000  0.000  0.000  0.000
|| cmpl1:_1:OS  0.000  0.000  0.000  0.000  0.000 -0.046  0.000  0.000  0.000  0.000  0.021
```

Critical follow-up:

```r
emmeans(
  m_nw_N400,
  ~ complexity*family_size | LP,
  at = list(LP = c(-1, 0, 1))
)
```

```
|| LP = -1:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large        0.0596 0.401 Inf    -0.727    0.8458
||  Simple     Large       -0.9885 0.401 Inf    -1.775   -0.2023
||  Complex    Small       -0.2403 0.401 Inf    -1.027    0.5459
||  Simple     Small       -0.6191 0.401 Inf    -1.405    0.1671
||
|| LP =  0:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large       -0.0523 0.317 Inf    -0.674    0.5696
||  Simple     Large       -0.7316 0.317 Inf    -1.353   -0.1098
||  Complex    Small       -0.3597 0.317 Inf    -0.982    0.2621
||  Simple     Small       -0.5991 0.317 Inf    -1.221    0.0227
||
|| LP =  1:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large       -0.1642 0.417 Inf    -0.981    0.6529
||  Simple     Large       -0.4748 0.417 Inf    -1.292    0.3423
||  Complex    Small       -0.4791 0.417 Inf    -1.296    0.3379
||  Simple     Small       -0.5791 0.417 Inf    -1.396    0.2379
||
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95
```

```r
emmeans(
  m_nw_N400,
  ~ complexity*family_size | OS,
  at = list(OS = c(-1, 0, 1))
)
```

```
|| OS = -1:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
```

```
||  Complex    Large       -0.3161 0.487 Inf    -1.271   0.63900
||  Simple     Large       -1.3376 0.487 Inf    -2.293  -0.38248
||  Complex    Small       -0.6291 0.487 Inf    -1.584   0.32600
||  Simple     Small       -1.0509 0.487 Inf    -2.006  -0.09582
||
|| OS =  0:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large       -0.0577 0.317 Inf    -0.680   0.56437
||  Simple     Large       -0.7679 0.317 Inf    -1.390  -0.14581
||  Complex    Small       -0.3650 0.317 Inf    -0.987   0.25703
||  Simple     Small       -0.6182 0.317 Inf    -1.240   0.00386
||
|| OS =  1:
||  complexity family_size  emmean    SE  df asymp.LCL asymp.UCL
||  Complex    Large        0.2007 0.465 Inf    -0.711   1.11240
||  Simple     Large       -0.1981 0.465 Inf    -1.110   0.71352
||  Complex    Small       -0.1009 0.465 Inf    -1.013   0.81073
||  Simple     Small       -0.1855 0.465 Inf    -1.097   0.72621
||
|| Degrees-of-freedom method: asymptotic
|| Confidence level used: 0.95
```

```r
contrast(
  emmeans(m_nw_N400, ~ complexity*family_size | LP,
          at = list(LP = c(-1, 1))),
  interaction = "pairwise"
)
```

```
|| LP = -1:
||  complexity_pairwise family_size_pairwise estimate    SE  df z.ratio p.value
||  Complex - Simple    Large - Small           0.669 0.155 Inf   4.310  <.0001
||
|| LP =  1:
||  complexity_pairwise family_size_pairwise estimate    SE  df z.ratio p.value
||  Complex - Simple    Large - Small           0.211 0.161 Inf   1.305  0.1918
||
|| Degrees-of-freedom method: asymptotic
```

0.4.2.1 N400 (nonwords): (Complex - Simple) difference scores as a function of continuous Language Proficiency (LP), averaged over Family Size .

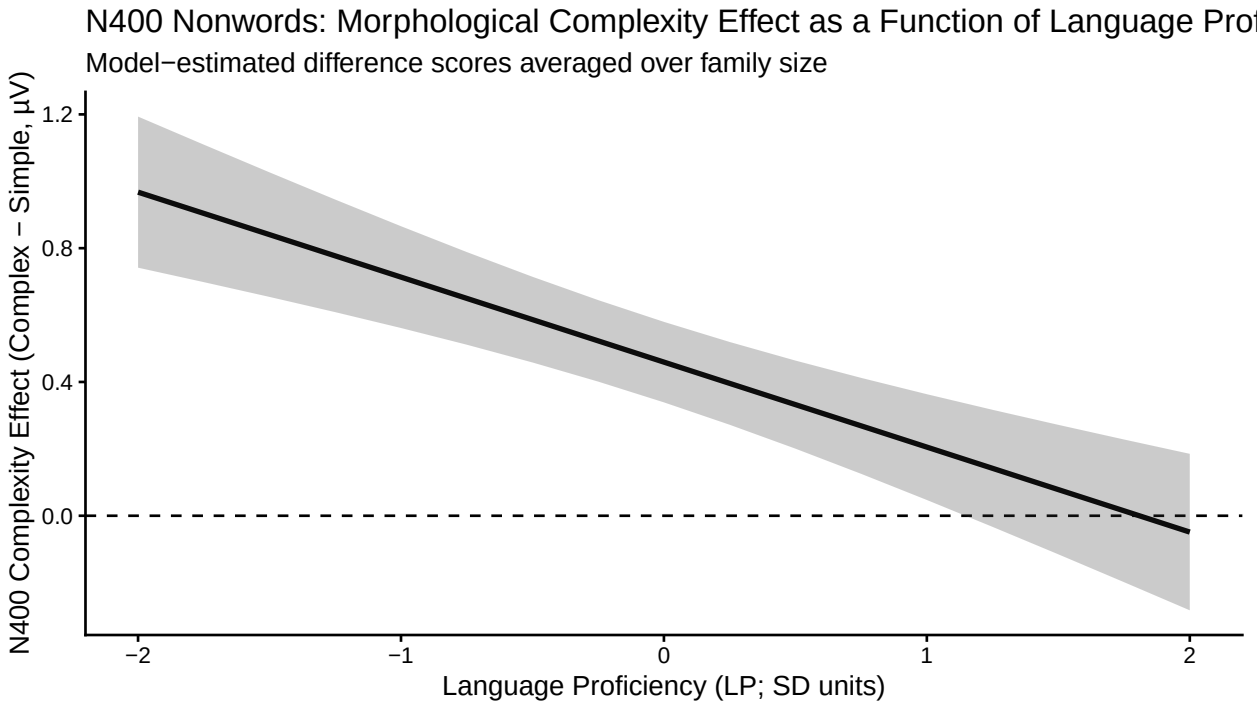
```r
# Generate complexity differences across LP

lp_grid <- seq(-2, 2, by = 0.25)

emm_lp <- emmeans(
  m_nw_N400,
  ~ complexity | LP,
  at = list(LP = lp_grid),
  weights = "proportional"
)

# Compute Complex - Simple difference at each LP
diff_lp <- contrast(emm_lp, method = "pairwise")
df_diff_lp <- as.data.frame(summary(diff_lp, infer = c(TRUE, TRUE)))


# Plot difference scores vs LP
ggplot(df_diff_lp, aes(x = LP, y = estimate)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_line(size = 1) +
  geom_ribbon(
    aes(ymin = asymp.LCL, ymax = asymp.UCL),
    alpha = 0.25
  ) +
  labs(
    x = "Language Proficiency (LP; SD units)",
    y = "N400 Complexity Effect (Complex - Simple, µV)",
    title = "N400 Nonwords: Morphological Complexity Effect as a Function of Language Proficiency",
    subtitle = "Model-estimated difference scores averaged over family size"
  ) +
  theme_classic()
```


N400 Nonwords: Morphological Complexity Effect as a Function of Language Prof
Model−estimated difference scores averaged over family size
N400 Complexity Effect (Complex − Simple, µV)
1.2
0.8
0.4
0.0
−2
−1
0
1
2
Language Proficiency (LP; SD units)